Muhammad Usama Tayyab

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience specializing in high-performance, cross-platform desktop development, embedded systems, and GUI design. Deep expertise in **Modern C++ (C++17/20), Qt (Widgets & QML)**, Linux environment engineering, and automation scripting. Proven track record architecting highly stable systems for **FDA-regulated medical devices** and cybersecurity applications. Adept at managing system architecture, real-time data streaming (TCP/UDP), performance profiling, and leading multi-functional engineering teams.

TECHNICAL SKILLS

Programming Languages	:	C++17/20 , C , Python
Frameworks and Libraries	:	PyQt , Qt 5.x / QML , G-Test
Scripting and Automation	:	Python , Shell/Bash , PowerShell
Source Code Management	:	Git , GitHub , BitBucket , Surround SCM
Project Management Tools	:	JIRA , Helix ALM , ClickUp
Build and Development	:	CMake , Make , GCC
Specializations	:	Data Structures & Algorithms, Object Oriented Programming, Design Patterns, Automation, CI/CD, Web Scraping, Data Analysis
Cloud and DevOps	:	AWS-S3 , Docker
Compliance Standards	:	MISRA C++ , Google C++ Style Guide, ISO C++ Code Guidelines

PROFESSIONAL EXPERIENCE

- **Senior Software Engineer (C++ developer)** **June 2021 - Present**
  Byonyks Pvt ltd.
  - Developed and maintained software for FDA-cleared Automated Peritoneal Dialysis (APD) machine using C++17, Qt framework, and embedded Linux.
  - Architectured inter-process communication system using Linux message queues for real-time data transfer between system components.
  - Collaborated with cross-functional teams (electrical, mechanical, QA engineers) to gather requirements and define new features for machine.
  - Conducted code reviews and mentored junior engineers in writing automated tests using Google Test framework, increasing code coverage to 95%.
  - Managed source control via GitHub and Surround SCM, tracked tasks using Helix & JIRA, and deployed periodic software releases through AWS.
  - Maintained comprehensive technical documentation for regulatory compliance and audits.
  - Reviewed large-scale therapy failure logs to identify root causes; developed C++, Python, and PowerShell scripts to automate log processing and analysis.
  - Resolved critical bugs in production, enhancing device reliability and increasing the success percentage of therapies from 60% to 91%.
- **Software Engineer (C++ developer)** **April 2020 - May 2021**
  Ebryx Ltd.
  - Developed cross-platform firewall desktop application using C, C++, Qt, and Python for macOS, Linux, and Windows.
  - Implemented socket programming and SDP protocol for TCP/UDP communication between controllers, gateways, and clients.
  - Integrated Single Packet Authentication (SPA) for secure data packet transmission.

- **Research Assistant at CSALT** **Sept 2019 - March 2020**
  Information Technology University
  - Contributed to Crime Investigation and Prevention Lab (CIPL) project focusing on audio analysis and keyword spotting.
  - Developed keyword spotting algorithms for audio utterance analysis.
  - Applied machine learning techniques to audio signal processing for automated detection systems.

ACADEMIC/HOBBY PROJECTS

- **Search Engine of Wikipedia articles:** Implemented a search engine of provided wikipedia articles using C++11 and STL containers (std::unordered_map and std::map) for data indexing and retrieval. Afterwards a comparative analysis was performed to measure efficiency of hash-based and tree-based data structures.
- **Disk Sort(External Sort):** Built an external sort engine in C++ that sorts files up to 100GB,exceeding available RAM; using chunk-based in-memory sorting and a std::priority_queue-based k-way merge, with custom test-data generation and independent correctness verification tooling.

EDUCATION & CERTIFICATIONS

- **Bachelor of Computer Science** **2015 - 2019**
  University of Central Punjab
- Data Analysis & Visualization with Python - Microsoft (Coursera)

  https://coursera.org/verify/PM6WXPQQR9RP
- Using Python to Access Web Data - University of Michigan (Coursera)

  https://coursera.org/verify/6RCFU7346HKM